

David Elijah Umaru

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SUMMARY

Computer Engineering senior (B.S., May 2026, GPA 3.94, Embedded Systems concentration) applying to graduate study in Computer Engineering. Builds real-time embedded control systems end-to-end — firmware, custom PCBs, and communication protocols — from a 1kHz motor-control loop on a quadrupedal robot to an ESP32-S3 robotic arm controller and campus HPC infrastructure.

EDUCATION

Colorado State University — Fort Collins, CO

May 2026

B.S. Computer Engineering, Minor in Computer Science — Concentration: Embedded Systems

GPA: 3.94/4.0 (Dean's List, 6 semesters)

EXPERIENCE

HPC Developer Intern

June 2025 – Present

Data Science Research Institute, Colorado State University

- Supported and troubleshoot high-availability Linux/HPC systems, improving computational reliability and reducing researcher downtime.
- Assisted with system validation and commissioning during a new HPC data-center rebuild, including configuration checks, node integration, and operational testing; authored internal technical documentation improving support efficiency ~50%.

Automation Intern

August 2022 – December 2022

Dept. of Microbiology, Immunology, and Pathology, Colorado State University

- Developed Python automation scripts using the Opentrons API for dilution, titration, and fluorescence workflows, increasing laboratory throughput ~50%.

IT Student Support & A/V Media Specialist

June 2023 – Present

College of Business, Colorado State University

- Resolved IT incidents for 275+ faculty and staff and deployed/imaged 150+ Windows systems for configuration consistency.

PROJECTS

RamBots — Real-Time Motor Control Firmware (Quadrupedal Robot)

Aug 2025 – Present

C++, Teensy 4.1, ODrive, CAN bus, interrupt-driven UART

- Rewrote the robot's C++ control firmware, replacing blocking serial I/O with interrupt-driven UART to raise the control loop from 20Hz to 1kHz, and built an inverse-kinematics solver with a modular Leg/Axis/ODrive class architecture enabling graceful degradation on single-motor faults.
- Parsed CAN bus telemetry (encoder position, current, fault codes) from six ODrive controllers; diagnosed an intermittent SPI cable fault via custom Python tooling, raising system uptime from 60% to 98%.
- Achieved <2° joint angle error and <50ms command-to-movement latency across 5+ live demonstrations with zero mid-presentation failures.

Custom PCB Design — FSAE Power Distribution & Safety Circuits

2024 – Present

KiCad, IPC-2221, JLCPCB fabrication

- Designed 3 PCBs (6-channel fused power distribution board, analog signal-conditioning board, fault-latch board) in KiCad, sizing copper trace widths to IPC-2221 for 40A continuous loads.
- Built an LM339 comparator circuit with a 555-timer fault-latch (500ms delay) for FSAE brake-plausibility compliance; validated 100% latch reliability across 200+ cycles and manufactured through JLCPCB at ~\$1,300 below commercial alternatives.

5-DOF Robotic Arm — Embedded Controller

Aug 2025 – Present

ESP32-S3, 4-layer PCB, JavaScript/HTML

- Designed a 4-layer PCB and ESP32-S3-based embedded controller for a 5-DOF robotic arm on a 6-member multidisciplinary senior design team.
- Wrote firmware for physical arm control and a JavaScript/HTML web interface for Wi-Fi/serial control.

Energy Management System — Edge ML Deployment

Sept 2024 – Dec 2024

Raspberry Pi, TensorFlow Lite, MQTT, Flask

- Deployed TensorFlow Lite on a Raspberry Pi for on-device inference over 20,000+ time-series data points, removing cloud dependency, and built an MQTT-based real-time data pipeline feeding a Flask dashboard at <100ms latency.

TECHNICAL SKILLS

Hardware & Embedded: PCB design (KiCad, IPC-2221), CAN bus, UART, SPI, Teensy 4.1, ESP32(-S3), Raspberry Pi, circuit wiring

Languages: C, C++, Python, VHDL, MATLAB, Java

Tools & Frameworks: Linux, Git/GitHub, Quartus Prime, TensorFlow Lite, VS Code, Onshape

RELEVANT COURSEWORK

Embedded Systems and Machine Learning, Digital System Design, VLSI Design, Digital Circuit Logic, Signals and Systems, Data Structures & Algorithms

LEADERSHIP & ACTIVITIES

- Electrical Low Voltage Lead, CSU Formula SAE Electric (2025–2026) — led a 7+ engineer electrical team; secured New Eagle sponsorship.
- Safety Officer, RamBots CSU Senior Design (2025–2026) — owned safety protocols and documentation for a 13-person multidisciplinary team.
- Ram Welcome Leader, CSU (Summer 2024) — mentored a group of 20 incoming students through campus transition.
- Member, National Society of Black Engineers (2023–2026) and ColorStack.

AWARDS & HONORS

Tesla Battery Sponsorship Recipient (2025) · ECE 202 Sophomore Competition Awardee · Fry Family Electrical Engineering Scholarship · Carl Wilsen Scholarship in ECE · Dean's List (6 semesters)